

Mathematics of Data Science - CA 4

Name: _____

Stieltjes transform 1 (Point masses)

Let μ be a probability measure consisting of one or more point masses. Recall the Stieltjes transform

$$S_\mu(z) = \int_{\mathbb{R}} \frac{1}{t-z} \mu(dt), \quad z \in \mathbb{C} \setminus \text{supp}(\mu).$$

We define δ_a to be a point mass at $a \in \mathbb{R}$.

Compute $S_\mu(z)$ explicitly in the following cases:

1. $\mu = (1 - \alpha) \delta_a + \alpha \delta_b$ with $a \neq b$ and $\alpha \in (0, 1)$.
2. $\mu = \sum_{k=1}^m p_k \delta_{a_k}$, where $a_1, \dots, a_m \in \mathbb{R}$ are distinct and $p_1, \dots, p_m > 0$ with $\sum_{k=1}^m p_k = 1$.

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4 pts

1. For $\mu = (1 - \alpha) \delta_a + \alpha \delta_b$,

$$S_\mu(z) = \int_{\mathbb{R}} \frac{1}{t-z} \mu(dt) = (1 - \alpha) \frac{1}{a-z} + \alpha \frac{1}{b-z},$$

valid for $z \notin \{a, b\}$.

2. For $\mu = \sum_{k=1}^m p_k \delta_{a_k}$,

$$S_\mu(z) = \int_{\mathbb{R}} \frac{1}{t-z} \mu(dt) = \sum_{k=1}^m p_k \int_{\mathbb{R}} \frac{1}{t-z} \delta_{a_k}(dt) = \sum_{k=1}^m \frac{p_k}{a_k - z},$$

valid for $z \notin \{a_1, \dots, a_m\}$.

Exercise 2 (Trace of a sample covariance matrix)

Let $\mathbf{X} \in \mathbb{R}^{p \times n}$ have columns $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$, and define the (uncentered) sample covariance

$$\hat{\Sigma} = \frac{1}{n} \mathbf{X} \mathbf{X}^\top \in \mathbb{R}^{p \times p}.$$

1. Derive a relation between $\text{tr}(\hat{\Sigma})$ and $\sum_{k=1}^n \|\mathbf{x}_k\|_2^2$.

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3 pts

2. Suppose now that the columns $\mathbf{x}_k$ are i.i.d. samples from a distribution with mean zero and covariance matrix Σ . What quantity does $\mathbb{E}[\frac{1}{p} \text{tr}(\hat{\Sigma})]$ estimate?

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3 pts

1)

$$\text{tr}(\hat{\Sigma}) = \frac{1}{n} \text{tr}(\mathbf{X} \mathbf{X}^\top) = \frac{1}{n} \text{tr}(\mathbf{X}^\top \mathbf{X}).$$

But $\mathbf{X}^\top \mathbf{X}$ has entries $(\mathbf{X}^\top \mathbf{X})_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$, so

$$\text{tr}(\hat{\Sigma}) = \frac{1}{n} \text{tr}(\mathbf{X}^\top \mathbf{X}) = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k^\top \mathbf{x}_k = \frac{1}{n} \sum_{k=1}^n \|\mathbf{x}_k\|_2^2.$$

- 2) For mean-zero data with covariance Σ , we have $\mathbb{E}[\hat{\Sigma}] = \Sigma$. Therefore,

$$\mathbb{E} \left[\frac{1}{p} \text{tr}(\hat{\Sigma}) \right] = \frac{1}{p} \text{tr}(\Sigma),$$

which is the average true variance across the p coordinates.